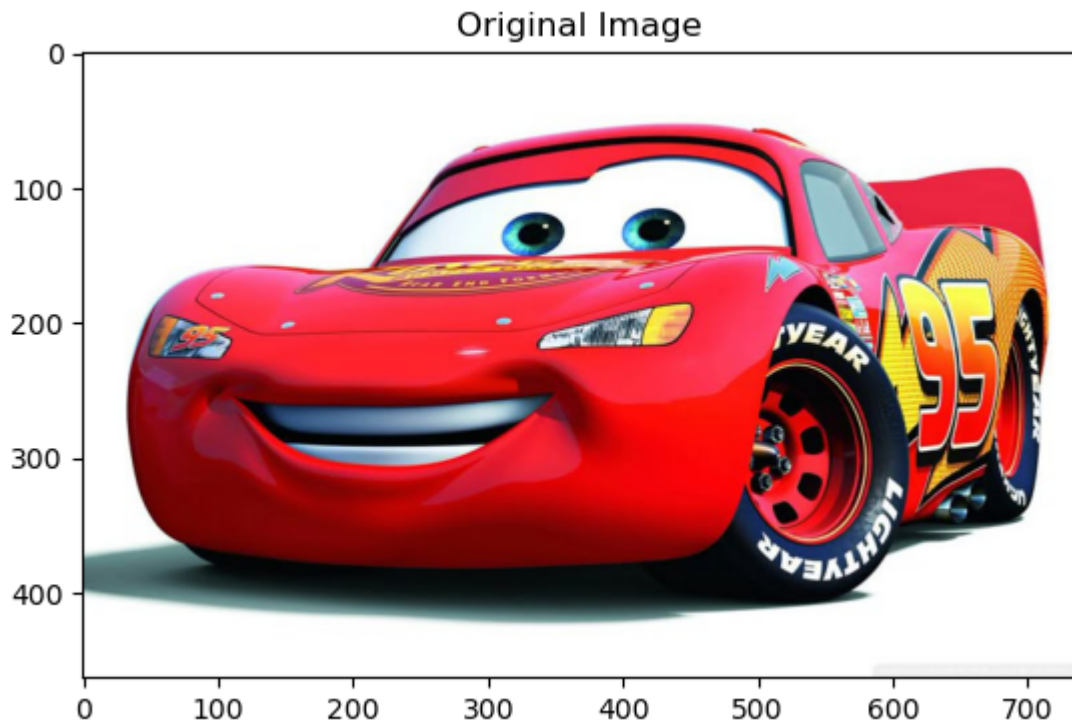


```
In [1]: import cv2
import matplotlib.pyplot as plt
```

```
In [2]: image = cv2.imread("download (2).jpg")
```

```
In [3]: plt.imshow(image[:,:,:-1])
plt.title("Original Image")
```

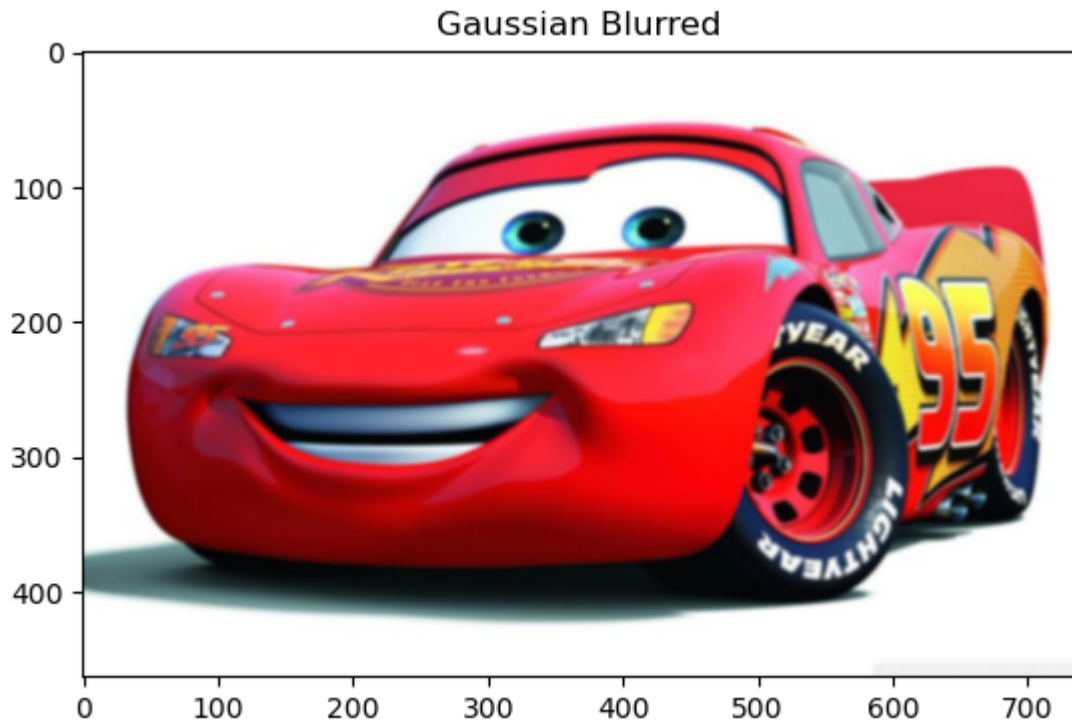
```
Out[3]: Text(0.5, 1.0, 'Original Image')
```



```
In [4]: # Apply Gaussian Blur to reduce noise
blurred = cv2.GaussianBlur(image, (5, 5), 1.4)
```

```
In [5]: plt.imshow(blurred[:,:,:-1])
plt.title("Gaussian Blurred")
```

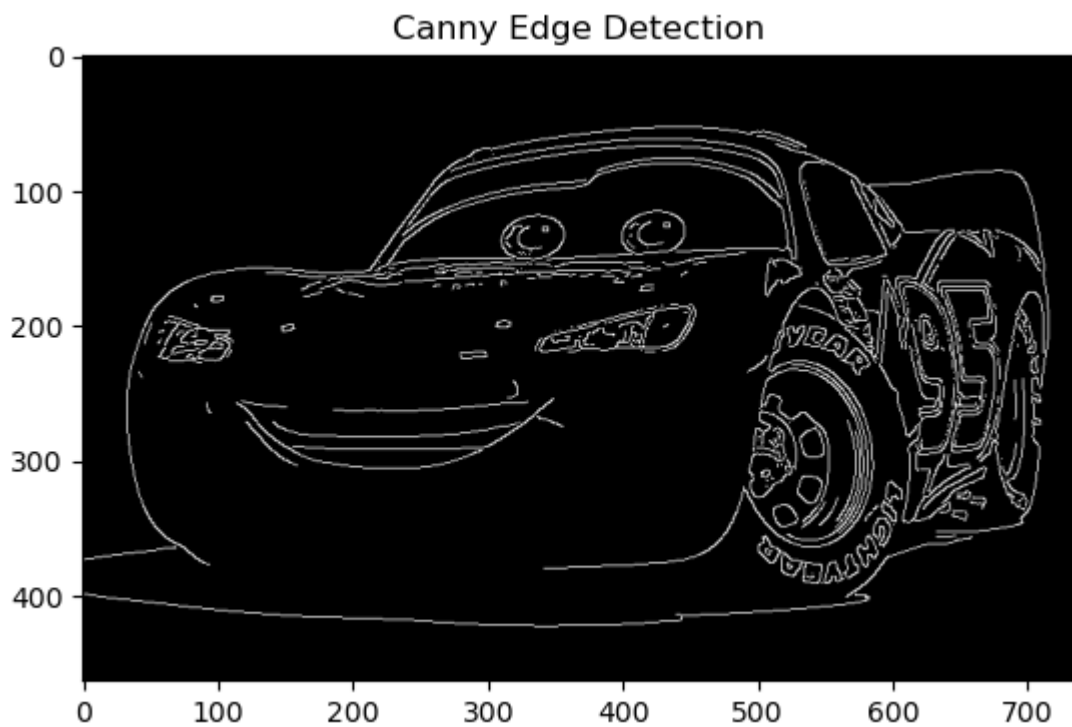
```
Out[5]: Text(0.5, 1.0, 'Gaussian Blurred')
```



```
In [6]: # Apply Canny Edge Detection
#Canny edge detection
edges = cv2.Canny(blurred, threshold1 = 180, threshold2 = 200)
```

```
In [7]: plt.imshow(edges, cmap="gray")
plt.title("Canny Edge Detection")
```

```
Out[7]: Text(0.5, 1.0, 'Canny Edge Detection')
```



```
In [8]: #Comparison of Images.
plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(image[:, :, :-1])
```

```
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(blurred[:, :, ::-1])
plt.title("Gaussian Blurred")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(edges, cmap="gray")
plt.title("Canny Edge Detection")
plt.axis("off")

plt.tight_layout()
plt.show()
```

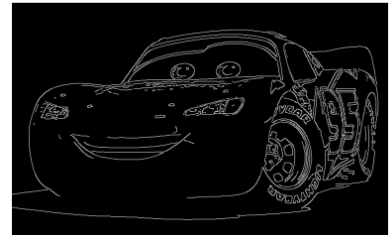
Original Image



Gaussian Blurred



Canny Edge Detection



In []: